

Notes on Hamiltonian Monte Carlo

Junsong Tang

May 21, 2026

1 Fundamentals of Hamiltonian Dynamics

1.1 Notations

- $\mathbf{q} = (q_1, \dots, q_n)$: generalized coordinates
- $\dot{\mathbf{q}} = (\dot{q}_1, \dots, \dot{q}_n)$: generalized velocity
- $\mathcal{L}(t, \mathbf{q}, \dot{\mathbf{q}}) = K - U = \frac{1}{2} \dot{\mathbf{q}}^\top M \dot{\mathbf{q}} - U(\mathbf{q})$: Lagrangian, where M is the mass matrix, K is kinetic energy and U is potential energy
- $\mathbf{p}$: generalized momentum, i.e., $\mathbf{p} = M \cdot \dot{\mathbf{q}}$

1.2 Euler–Lagrange Equation

Generalized version: given a function $\mathbf{y}(x)$ and its derivative: $\mathbf{y}'(x)$ and $f(x, \mathbf{y}, \mathbf{y}')$. Define the functional:

$$F[\mathbf{y}] = \int_a^b f(x, \mathbf{y}, \mathbf{y}') dx \quad (1)$$

As the topic in calculus of variation, we want the functional $F[\mathbf{y}]$ to obtain its local min for some function $\mathbf{y}$. Perturbate $\mathbf{y}(x)$ with $\mathbf{y}(x) + \varepsilon \mathbf{u}(x)$ for any function $\mathbf{u}$ and $\varepsilon \in \mathbb{R}$ with small enough $|\varepsilon|$. and put

$$g(\varepsilon) = \int_a^b f(x, \mathbf{y} + \varepsilon \mathbf{u}, (\mathbf{y} + \varepsilon \mathbf{u})') dx$$

To make $F[\mathbf{y}]$ obtaining its local minimum for some $\mathbf{y}$ satisfying the initial conditions: $\mathbf{y}(a)$ and $\mathbf{y}(b)$ being fixed, i.e., $\mathbf{u}(a) = \mathbf{u}(b) = 0$, it is equivalent to require $g'(0) = 0$, hence:

$$\begin{aligned} g'(0) &= \frac{d}{d\varepsilon} \left(\int_a^b f(x, \mathbf{y} + \varepsilon \mathbf{u}, (\mathbf{y} + \varepsilon \mathbf{u})') dx \right) \\ &= \int_a^b \left(\frac{\partial f}{\partial \mathbf{y}} \mathbf{u} + \frac{\partial f}{\partial \mathbf{y}'} \mathbf{u}' \right) dx \quad \text{By total derivative} \\ &= \int_a^b \mathbf{u} \frac{\partial f}{\partial \mathbf{y}} dx + \frac{\partial f}{\partial \mathbf{y}'} \mathbf{u} \Big|_a^b - \int_a^b \mathbf{u} \frac{d}{dx} \left(\frac{\partial f}{\partial \mathbf{y}'} \right) dx \quad \text{integration by parts} \\ &= \int_a^b \mathbf{u} \left(\frac{\partial f}{\partial \mathbf{y}} - \frac{d}{dx} \frac{\partial f}{\partial \mathbf{y}'} \right) dx = 0 \quad \text{By initial condition: } \mathbf{u}(a) = \mathbf{u}(b) = 0 \end{aligned}$$

By the fundamental lemma of calculus of variation, since $\mathbf{u}(x)$ is arbitrary, hence we must have:

$$\frac{\partial f}{\partial \mathbf{y}} - \frac{d}{dx} \frac{\partial f}{\partial \mathbf{y}'} = 0 \quad (2)$$

Equation (2) is called the **Euler–Lagrangian Equation** in calculus of variation, and for $\mathbf{y}$ satisfying equation (2) is a sufficient condition for $F[\mathbf{y}]$ to have its local minimum.

In a dynamical system, with Lagrangian $\mathcal{L}(t, \mathbf{q}, \dot{\mathbf{q}})$, we replace f with $\mathcal{L}$, $\mathbf{y}$ with generalized coordiantes $\mathbf{q}$, and x with time t , we get the **Lagrangian Equation** for dynamical system:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{q}} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} = 0 \quad (3)$$

1.3 Legendre Transform

If $f : \mathcal{X} \rightarrow \mathbb{R}$ is convex, then $\forall t \in \mathcal{X}$, then $f(t) \geq pt + b$, where $p = f'(x)$ and $b = f(x) - px$. This implies that:

$$pt - f(t) \leq px - f(x), \forall t \in \mathcal{X}$$

Then the one-dimensional Legendre transform is defined as:

$$f^*(p) = px - f(x) = \sup\{pt - f(t) : t \in \mathcal{X}\}$$

If $\mathcal{X} \subset \mathbb{R}^n$, and $\mathbf{p} = \nabla f = \frac{\partial f}{\partial \mathbf{x}}$, then the generalized Legendre tranform is defined as:

$$f^*(\mathbf{p}) = \mathbf{p} \cdot \mathbf{x} - f(\mathbf{x}) = \sup\{\mathbf{p} \cdot \mathbf{t} - f(\mathbf{t}) : \mathbf{t} \in \mathcal{X}\}$$

where $\mathbf{p} \cdot \mathbf{x} = \mathbf{p}^\top \mathbf{x}$ representing the inner product.

Note that f^* is also a convex function, since given a $t \in \mathcal{X}$, the map $p \mapsto pt - f(t)$ is linear, hence the supremum of the family of linear maps: $\{p \mapsto pt - f(t)\}_{t \in \mathcal{X}}$ gives an envelope, which is f^* . On the other hand, one can show:

$$f(x) = \sup\{px - f^*(p) : p \in \mathcal{X}^*\}$$

which means that $f(x)$ is the envelope of the family of linear maps: $\{x \mapsto px - f^*(p)\}_{p \in \mathcal{X}^*}$.

1.4 Hamiltonian Equation

Hamiltonian is the Legendre transform of Lagrangian, hence we have:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \mathbf{p} \cdot \dot{\mathbf{q}} - \mathcal{L}(t, \mathbf{q}, \dot{\mathbf{q}}) \quad (4)$$

Since $\mathbf{p} \cdot \dot{\mathbf{q}} = 2K$, so $\mathcal{H}(\mathbf{q}, \mathbf{p}) = 2K - (K - U) = K + U$ and hence Hamiltonian can be interpreted as the total energy of a dynamical system. So (4) can also be written as:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2} \dot{\mathbf{q}}^\top M \dot{\mathbf{q}} + U(\mathbf{q}) = \frac{1}{2} \mathbf{p}^\top M^{-1} \mathbf{p} + U(\mathbf{q}) \quad (5)$$

We can derive Hamiltonian equations from Lagrange eqaution in (3).

Note that by $\mathcal{L}(t, \mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^\top M \dot{\mathbf{q}} - U(\mathbf{q})$, we have:

$$\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} = M \cdot \dot{\mathbf{q}} = \mathbf{p} \quad \text{and by (3)} \quad \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} = \dot{\mathbf{p}}$$

so with (4) and (5), we have:

$$\frac{\partial \mathcal{H}}{\partial \mathbf{p}} = \dot{\mathbf{q}} \quad \text{and} \quad \frac{\partial \mathcal{H}}{\partial \mathbf{p}} = M^{-1} \mathbf{p} \quad (6)$$

$$\frac{\partial \mathcal{H}}{\partial \mathbf{q}} = -\frac{\partial \mathcal{L}}{\partial \mathbf{q}} = -\dot{\mathbf{p}} \quad \text{and} \quad \frac{\partial \mathcal{H}}{\partial \mathbf{q}} = \nabla_{\mathbf{q}} U(\mathbf{q}) \quad (7)$$

(6) and (7) are called **Hamiltonian equations**

Or equivalently, we can take the total differentiation on both sides of (4):

$$\begin{aligned}
d\mathcal{H} &= d(\mathbf{p} \cdot \dot{\mathbf{q}}) - d\mathcal{L}(t, \mathbf{q}, \dot{\mathbf{q}}) \\
&= \dot{\mathbf{q}} \cdot d\mathbf{p} + \mathbf{p} \cdot d\dot{\mathbf{q}} - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} d\mathbf{q} - \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} d\dot{\mathbf{q}} - \frac{\partial \mathcal{L}}{\partial t} dt \\
&= \dot{\mathbf{q}} \cdot d\mathbf{p} - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} d\mathbf{q} - \frac{\partial \mathcal{L}}{\partial t} dt \\
&= \frac{\partial \mathcal{H}}{\partial \mathbf{p}} d\mathbf{p} + \frac{\partial \mathcal{H}}{\partial \mathbf{q}} d\mathbf{q} + \frac{\partial \mathcal{H}}{\partial t} dt
\end{aligned}$$

Hence we can correspond the coefficients to get (6) and (7).

By combining (6) and (7), we obtain another form of Hamiltonian equations:

$$\dot{\mathbf{q}} = M^{-1}\mathbf{p} \quad (8)$$

$$\dot{\mathbf{p}} = -\nabla_{\mathbf{q}} U(\mathbf{q}) \quad (9)$$

which will be used in the implementation of Hamiltonian Monte Carlo.

2 Hamiltonian Monte Carlo

2.1 Leapfrog integrator

The numerical approximation of Hamiltonian dynamics, which is a system of ODE: 8, 9, can be obtained by integrators. Particularly, the leapfrog integrator is the most preferred.

$$\begin{aligned}
p_{t+\varepsilon/2} &= p_t - \frac{\varepsilon}{2} \nabla U(x_t) \quad (\text{kick}) \\
x_{t+\varepsilon} &= x_t + \varepsilon M^{-1} p_{t+\varepsilon/2} \quad (\text{drift}) \\
p_{t+\varepsilon} &= p_{t+\varepsilon/2} - \frac{\varepsilon}{2} \nabla U(x_{t+\varepsilon}) \quad (\text{kick})
\end{aligned}$$

If we expand the above, we could get the approximate state: $(\hat{x}_{t+\varepsilon}, \hat{p}_{t+\varepsilon})$ at time $t + \varepsilon$ from time t :

$$\begin{aligned}
x_{t+\varepsilon} &= x_t + \varepsilon M^{-1} p_t - \frac{\varepsilon^2}{2} M^{-1} \nabla U(x_t) \\
p_{t+\varepsilon} &= p_t - \varepsilon \nabla U(x_t) - \frac{\varepsilon}{2} (\nabla U(x_{t+\varepsilon}) - \nabla U(x_t)) \\
&= p_t - \varepsilon \nabla U(x_t) - \frac{\varepsilon^2}{2} \nabla^2 U(x_t) M^{-1} p_t + O(\varepsilon^3)
\end{aligned}$$

This is consistent with second-order Taylor expansion:

$$\begin{aligned}
x_{t+\varepsilon} &= x_t + \varepsilon M^{-1} p_t - \frac{\varepsilon^2}{2} M^{-1} \nabla U(x_t) + O(\varepsilon^3) \\
p_{t+\varepsilon} &= p_t - \varepsilon \nabla U(x_t) - \frac{\varepsilon^2}{2} \nabla^2 U(x_t) M^{-1} p_t + O(\varepsilon^3)
\end{aligned}$$

Since the energy is conserved in Hamiltonian dynamics, so it is natural to consider the Hamiltonian: $\mathcal{H}(x, p) = C$ as a level set in phase space: $\mathbb{R}^d \times \mathbb{R}^d$, and the tangent space at state (x_t, p_t) is naturally:

$$\text{span}\left\{\left(\frac{dx}{dt}, \frac{dp}{dt}\right)^\top\right\} = \text{span}\left\{\left(M^{-1}p_t, -\nabla U(x_t)\right)^\top\right\}$$

and the normal space is:

$$\text{span}\left\{\left(\frac{\partial \mathcal{H}}{\partial x}, \frac{\partial \mathcal{H}}{\partial p}\right)^\top\right\} = \text{span}\left\{\left(\nabla U(x_t), M^{-1}p_t\right)^\top\right\}$$

Consider starting from the current state (x_t, p_t) , we propose some tangent move: $\varepsilon(M^{-1}p_t, -\nabla U(x_t))$, and the point being shot off the manifold:

$$(\tilde{x}_{t+\varepsilon}, \tilde{p}_{t+\varepsilon}) = \left(x_t + \varepsilon M^{-1}p_t, p_t - \varepsilon \nabla U(x_t)\right)$$

Note that the difference between $(\tilde{x}_{t+\varepsilon}, \tilde{p}_{t+\varepsilon})$ and the leapfrog solution of the state $(\hat{x}_{t+\varepsilon}, \hat{p}_{t+\varepsilon})$ at time $t + \varepsilon$ is:

$$\Delta = -\frac{\varepsilon^2}{2} \left(M^{-1} \nabla U(x_t), \nabla^2 U(x_t) M^{-1} p_t \right)$$

One question is: is this vector in the normal space of (x_t, p_t) . The answer is no in general. Because we need to enforce the condition:

$$\left(M^{-1} \nabla U(x_t), \nabla^2 U(x_t) M^{-1} p_t \right)^\top = c \left(\nabla U(x_t), M^{-1} p_t \right)^\top$$

for some scalar c , which requires that $\nabla U(x_t)$ is the eigenvector of M^{-1} and $M^{-1}p_t$ is the eigenvector of $\nabla^2 U(x_t)$ with the same eigenvalue c .

However, in some special cases, that can happen. Consider the simple harmonic oscillator with $M = m$ and spring constant $k = \frac{1}{m}$, so $U(x) = \frac{k}{2}x^2$, then in this case, $\Delta \propto (kx/m, kp/m)$ and the normal vector is $\propto (kx, p/m)$, therefore by setting $c = 1/m$ the requirements can be satisfied.